WORKED SOLUTION

STEP II 2014, Question 5

Pure • Homogeneous Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q5

The key idea

Both right-hand sides are ratios of linear expressions. In (i) every term has degree one, so the gradient depends only on y/x, and putting $y = xu$ turns the equation into a separable one. In (ii) the constant terms spoil this, but they can be removed by moving the origin to the point where the two lines in the numerator and denominator meet; after that it is (i) all over again.

Solution

If $y = xu$, the product rule gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = u + x\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Part (i)

Dividing the top and bottom of the right-hand side by x, the equation becomes

$$u + x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2u+1}{u-2}, \qquad \text{so} \qquad x\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2u+1-u(u-2)}{u-2} = \frac{1+4u-u^2}{u-2}.$$

Separating the variables,

$$\int \frac{u-2}{1+4u-u^2}\,\mathrm{d}u = \int \frac{1}{x}\,\mathrm{d}x.$$

? **Ask yourself** How do we integrate the left-hand side without partial fractions?

Differentiate the denominator: $\frac{\mathrm{d}}{\mathrm{d}u}(1+4u-u^2) = 4-2u = -2(u-2)$. So the numerator is $-\frac{1}{2}$ times the derivative of the denominator, and

$$-\tfrac{1}{2}\ln\left|1+4u-u^2\right| = \ln|x| + C.$$

Multiplying by -2 and exponentiating, $x^2\,|1+4u-u^2| = A$ for a positive constant A, that is, $x^2(1+4u-u^2) = K$ for some constant K (the sign is fixed along the curve). Now substitute $u = y/x$, so that $x^2(1+4y/x-y^2/x^2) = x^2+4xy-y^2$:

$$x^2+4xy-y^2 = K.$$

At $(1,1)$ we get $K = 1+4-1 = 4$, so the solution curve is

$$\boxed{x^2+4xy-y^2 = 4}$$

(At $(1,1)$ we have $u = 1$, where $1+4u-u^2 = 4 \neq 0$ and $u \neq 2$, so nothing we divided by vanishes at the starting point.)

Part (ii)

The lines $x-2y-4 = 0$ and $2x+y-3 = 0$ meet where $y = 3-2x$ and $x-2(3-2x)-4 = 0$, i.e. at $(2,-1)$. So take $a = 2$, $b = -1$: $x = X+2$, $y = Y-1$. Then $\mathrm{d}y/\mathrm{d}x = \mathrm{d}Y/\mathrm{d}X$, and

$$\begin{aligned} x-2y-4 &= (X+2)-2(Y-1)-4 = X-2Y, \\ 2x+y-3 &= 2(X+2)+(Y-1)-3 = 2X+Y, \end{aligned}$$

so the constants disappear:

$$\frac{\mathrm{d}Y}{\mathrm{d}X} = \frac{X-2Y}{2X+Y}.$$

This has the same shape as (i), so put $Y = XV$:

$$V+X\frac{\mathrm{d}V}{\mathrm{d}X} = \frac{1-2V}{2+V} \quad\Longrightarrow\quad X\frac{\mathrm{d}V}{\mathrm{d}X} = \frac{1-2V-V(2+V)}{2+V} = \frac{1-4V-V^2}{2+V}.$$

Again the numerator is a multiple of the derivative of the denominator, since $\frac{\mathrm{d}}{\mathrm{d}V}(1-4V-V^2) = -2(2+V)$. Hence

$$\int \frac{2+V}{1-4V-V^2}\,\mathrm{d}V = \int \frac{1}{X}\,\mathrm{d}X \quad\Longrightarrow\quad -\tfrac{1}{2}\ln\left|1-4V-V^2\right| = \ln|X|+C,$$

and as before $X^2(1-4V-V^2) = K$, i.e.

$$X^2-4XY-Y^2 = K.$$

The point $(1,1)$ corresponds to $X = -1$, $Y = 2$, giving $K = 1-4(-1)(2)-4 = 5$. Returning to x and y,

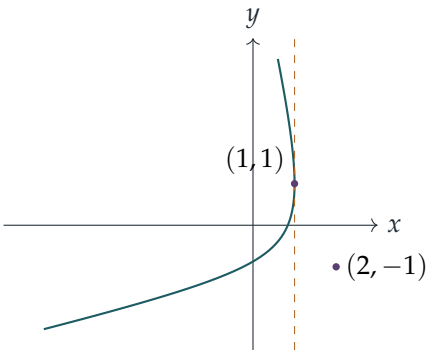
$$(x-2)^2-4(x-2)(y+1)-(y+1)^2 = 5,$$

and expanding,

$$\begin{aligned} &(x^2-4x+4)-4(xy+x-2y-2)-(y^2+2y+1) = 5 \\ \Longleftrightarrow\quad &\boxed{x^2-4xy-y^2-8x+6y+6 = 0} \end{aligned}$$

Ask yourself At $(1,1)$ the denominator $2x+y-3$ is zero. Does the solution still make sense?

Yes: the gradient is infinite there, so the curve has a vertical tangent at $(1,1)$. The implicit equation describes the curve perfectly well; it just isn't the graph of a function $y(x)$ through that point. Equivalently, x is a smooth function of y. Solving the quadratic for x gives $x = 2y+4-\sqrt{5y^2+10y+10}$ on the branch through $(1,1)$, which is defined for every y since $5y^2+10y+10 = 5(y+1)^2+5 > 0$, and which has its largest value $x=1$ at $y=1$.



The figure shows the branch through $(1,1)$ (teal), touching the vertical line $x=1$ (dashed) there; $(2,-1)$ is the centre of the hyperbola.

Checks

- Differentiate the answers implicitly. In (i), $2x+4y+(4x-2y)y' = 0$ gives $y' = \dfrac{x+2y}{y-2x}$, as required. In (ii), $2x-4y-8+(6-4x-2y)y' = 0$ gives $y' = \dfrac{x-2y-4}{2x+y-3}$.
- Both curves pass through $(1,1)$: $1+4-1=4$ and $1-4-1-8+6+6=0$.
- Stepping each differential equation numerically away from $(1,1)$ by computer, the left-hand sides of the boxed equations stayed equal to 4 and 0 to about 10^{-13}. In (i) the gradient at $(1,1)$ is -3.

Common slips

- Forgetting the u in $\mathrm{d}y/\mathrm{d}x = u + x\,\mathrm{d}u/\mathrm{d}x$, which leaves a wrong separable equation.
- Applying the condition in the wrong coordinates in (ii): the point $(1,1)$ becomes $(X,Y) = (-1,2)$, not $(1,1)$.
- Reaching for partial fractions in u: the roots of $1+4u-u^2$ are $2\pm\sqrt{5}$, which is messy, whereas the integrand is already of the form f'/f.

Beyond STEP

Equations whose right-hand side depends only on y/x are called homogeneous, and the shift of origin in (ii) is the standard trick for ratios of linear functions. Both equations here are also exact: writing (i) as $(2y + x)\,\mathrm{d}x - (y - 2x)\,\mathrm{d}y = 0$, the two coefficients satisfy the cross-derivative test, and the conserved quantity $x^2 + 4xy - y^2$ can be read off directly. Exact equations, integrating factors and phase portraits of linear systems (which is what the hyperbolae really are) are covered in Boyce and DiPrima, *Elementary Differential Equations*.

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